

DESIDIET: An Artifact-Grounded Systems Evaluation of Culturally Tailored Knowledge Graphs, Deterministic Portion Optimization, and Plan Verification for South Asian Nutrition

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Abstract

Personalized clinical dietary decision support in South Asia faces severe public health challenges driven by the dual burden of non-communicable diseases (NCDs) and widespread micronutrient malnutrition. In Bangladesh, where NCDs account for 67% of annual mortality, ungrounded generative large language models (LLMs) frequently hallucinate food portion sizes, omit cultural cooking staples such as added salt and cooking oil, violate critical clinical contraindications, and output culturally bizarre food combinations. Furthermore, prevailing GraphRAG methods rely on scale-invariant cosine similarity, which evaluates nutrient vector orientation rather than physical caloric and quantitative adequacy. This paper presents an artifact-grounded systems and engineering evaluation of DESIDIET, an open-source, culturally tailored clinical nutrition framework for Bangladesh. DESIDIET anchors dietary guidance in four deterministic software artifacts: (1) a canonical knowledge graph synthesizing 582 regional foods and clinical ontologies reconciled from the Bangladesh Food Composition Table (FCT 2014), the peer-reviewed GraphRAG dataset by Dindukurthi *et al.* (*Frontiers in Artificial Intelligence*, 2026), and the *National Dietary Guidelines for Bangladesh 2025*, validated against thermodynamic mass-balance bounds (95–105 g proximate sum); (2) a culturally curated Food Compatibility Matrix (575 items, 2,328 meal-slot edges, 1,150 pairing affinities) enforcing authentic culinary roles (staple, main, side, supplementary) to eliminate bizarre combinations and deliver meaningful food suggestions; (3) a clinical reference engine codifying national life-stage Recommended Dietary Allowances (RDA); and (4) a portion-constrained linear optimization engine that computes exact gram weights across daily meal slots while enforcing strict caps on spices, salt (≤ 5 g), and cooking oil (15–30 g). Candidate plans from bilingual LLMs are intercepted by an independent, deterministic Plan Verifier that audits ingredient-level nutrient sums, catches clinical contraindications, and executes a bounded repair loop for caloric drift exceeding 5%. Evaluating DESIDIET across 13 cross-lingual clinical scenarios demonstrates that the system completely eliminates clinical contraindications (from 0.62 to 0.00 violations per plan), constrains caloric drift to 2.45% ($p < 0.001$), and elevates culinary realism to 9.27/10, providing a verifiable and reproducible foundation for AI-assisted dietetics.

Keywords: clinical decision support, personalized nutrition, knowledge graph, portion optimization, food compatibility, deterministic verification, Bangladesh, South Asian diet

1. Introduction

Chronic non-communicable diseases (NCDs) have emerged as the primary cause of premature mortality and long-term morbidity across South Asia. In Bangladesh, cardiovascular diseases, Type 2 diabetes mellitus (T2D), chronic obstructive pulmonary disease, and chronic kidney disease (CKD) account for 67% of all annual deaths [1]. Concurrently, micronutrient malnutrition remains endemic, with iron-deficiency anemia affecting an estimated 40% of women of reproductive age and 33% of children under five [2, 3].

1.1. Pathophysiological & Epidemiological Context

The clinical management of these dual burdens is compounded by the distinct South Asian “thin-fat” metabolic phenotype [4]. Compared to Caucasian reference cohorts of identical body mass index (BMI), South Asians exhibit significantly higher percentages of body fat, elevated visceral adiposity, and pronounced insulin resistance at lower adiposity levels [4]. Consequently, the World Health Organization (WHO) has established population-specific clinical cutoffs for Asian populations (overweight ≥ 23.0 kg/m², obesity ≥ 27.5 kg/m²). This metabolic vulnerability is severely exacerbated by the traditional Bangladeshi dietary structure:

1. **Carbohydrate Density and Glycemic Surge:** Polished parboiled rice (*mota chal* and *miniket*) provides 70–75% of daily caloric intake in rural and semi-urban

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Bangladesh, generating high post-prandial glycemic excursions that accelerate beta-cell exhaustion.

2. **High-Phytate Chelation:** Non-heme iron and zinc absorption is severely impaired by unfermented pulses and whole grains with high myo-inositol hexakisphosphate (phytate) levels, where phytate-to-iron molar ratios regularly exceed 15 : 1 (far surpassing the inhibitory threshold of 1 : 1) [5].
3. **Unmonitored Cooking Mediums and Sodium:** Traditional curry preparations rely heavily on cooking fats (mustard oil, refined soybean oil) and free salt. Sodium intake in Bangladesh averages 11.5 g/day of salt—more than double the WHO recommended limit of 5.0 g/day [6]. Mustard oil, while containing beneficial α -linolenic acid (ω -3), contains $\approx$ 40% erucic acid (22 : 1 ω -9), requiring bounded thermal usage.

1.2. Clinical Pitfalls of Generative Models

Despite the urgent public health demand, access to professional clinical dietary counseling in Bangladesh is severely constrained, with fewer than one qualified dietitian per 100,000 citizens. Automated digital decision-support systems have been widely proposed to democratize nutritional counseling. While recent advances in large language models (LLMs) enable fluid, bilingual conversational interaction in English and Bengali, general-purpose generative models exhibit fatal deficiencies when deployed in clinical dietetics [7, 8]:

1. **Portion Hallucination and Unbounded Arithmetic:** LLMs routinely generate qualitative food recommendations (*e.g.*, “a bowl of rice with fish curry”) devoid of explicit gram portions, or output nutritional summaries that diverge by 20–40% from the physical sum of the constituent foods [7].
2. **Omission of Cultural Cooking Mediums:** Generative models almost universally omit cooking oils and added salt from their internal arithmetic, producing a systematic underestimation of daily energy intake by 200–450 kcal and failing sodium tracking in hypertensive patients.
3. **Failure of Hard Clinical Contraindications:** Because probabilistic token predictors lack deterministic constraint enforcement, they frequently recommend high-potassium foods (*e.g.*, spinach, lentils, pumpkin) to Stage 3/4 CKD patients or simple sugars to diabetics [9].
4. **Culinary Absurdity and Bizarre Food Combinations:** Pure mathematical optimizers and generic LLMs lack culinary ontology, frequently prescribing bizarre, unpalatable food combinations (*e.g.*, prescribing beef liver or heavy fish curry for breakfast, recommending raw lentils without cereal carriers, combining five uncoordinated pulses in one meal, or pairing milk

tea with oily curries) that violate cultural norms and destroy patient adherence.

1.3. Critique of GraphRAG Cosine Heuristics

To mitigate generative hallucinations, Knowledge Graphs (KGs) and Graph-Retrieved Augmented Generation (GraphRAG) have been introduced [10, 11]. A prominent recent baseline, Dindukurthi *et al.* [12], proposed a GraphRAG framework for Indian diets using Neo4j and cosine similarity over nutrient vectors. However, an empirical inspection of angular cosine ranking reveals a fatal mathematical limitation: *cosine similarity evaluates vector direction, not nutritional adequacy*. Because cosine similarity is scale-invariant:

$$\text{Cosine}(\alpha \mathbf{V}, \mathbf{1}) = \text{Cosine}(\mathbf{V}, \mathbf{1}) \quad \forall \alpha > 0 \quad (1)$$

a food vector providing 5% of daily nutrient targets and one providing 90% score identically against an all-ones target vector. Consequently, cosine ranking cannot determine gram portions, cannot solve multi-food combinations, and permits pathological recommendations (*e.g.*, prescribing 100 g of cumin or turmeric to satisfy an iron deficit). Furthermore, prior architectures lack an independent post-generation verifier, allowing ungrounded text to reach users unchecked.

1.4. Scope and Research Questions

This paper presents an **artifact-grounded systems and engineering evaluation** of DESIDIET (Figure 1). We do not claim prospective clinical trial efficacy in improving patient biomarkers (such as HbA1c or glomerular filtration rate), which requires formal Institutional Review Board (IRB) approved medical trials. Instead, we frame our contributions around verified software artifacts, mathematical proofs of algorithmic limitations, and empirical verification audits, addressing four research questions:

- **RQ1 (Data Integrity):** How can population-specific food tables be reconciled to guarantee thermodynamic mass balance and clinical safety?
- **RQ2 (Algorithmic Rigor & Culinary Realism):** Why does angular cosine retrieval fail nutritional adequacy, and how can linear portion optimization coupled with a graph-based Food Compatibility Matrix guarantee exact daily requirements while generating meaningful, authentic food suggestions?
- **RQ3 (Deterministic Verification):** Can an independent post-generation verification oracle bound generative caloric drift to $< 5\%$ and eliminate clinical contraindications?
- **RQ4 (Systems Performance):** What are the latency and cross-lingual robustness characteristics of the verified architecture under real-world deployment?

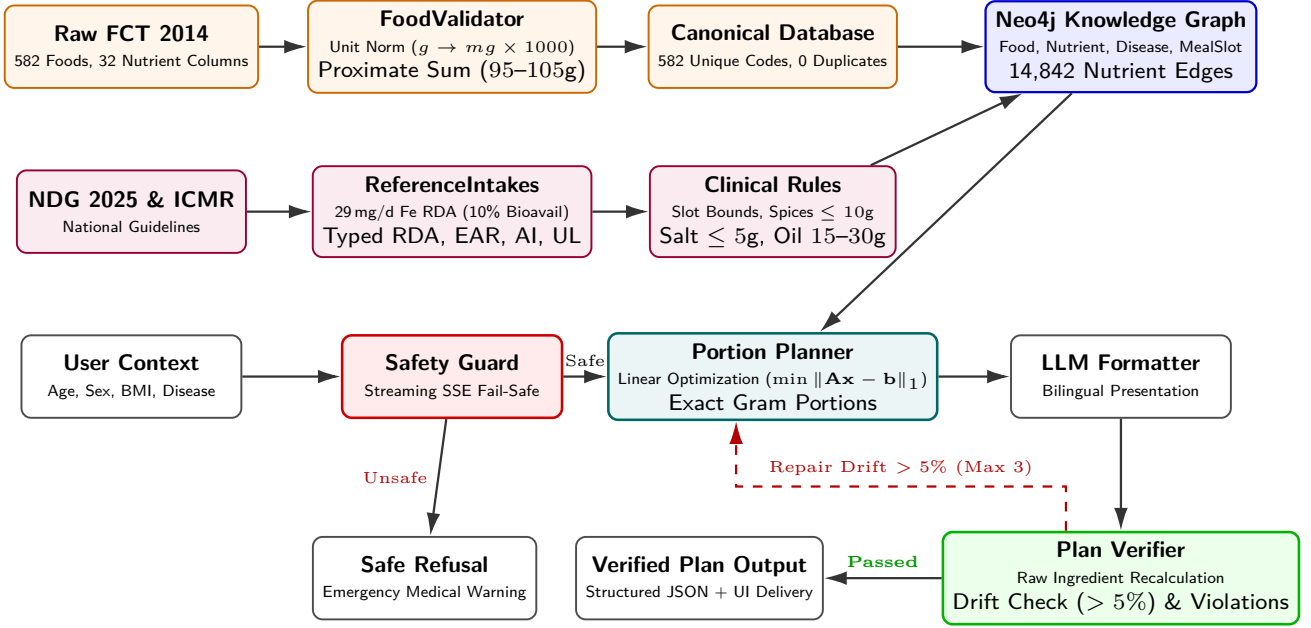


Figure 1: **Data and serving evidence flow in DesiDiet.** The architecture enforces an unbroken chain of custody: raw food composition data is standardized and validated against thermodynamic proximate sum envelopes before graph ingestion; clinical rules are typed and adjudicated; exact portions are computed via linear optimization; and candidate plans must pass independent post-generation verification before user delivery.

2. Related Work & Theoretical Foundations

2.1. Computational Diet Planning and Optimization

Mathematical diet formulation dates to the classical Stigler cost-of-subsistence problem [13] and Dantzig’s linear programming (LP) simplex algorithm [14]. While linear programming guarantees exact mathematical optimality, early single-objective models produced culturally intolerable solutions (*e.g.*, prescribing kilograms of pure flour or lard) [13]. Modern nutrition informatics has introduced mixed-integer linear programming (MILP) with palatability and variety constraints [15, 16]. However, traditional mathematical solvers operate in isolated spreadsheets, lacking the conversational interfaces, natural language explanations, and bilingual comprehension necessary for broader patient adherence [7].

2.2. Knowledge Graphs and GraphRAG in Medicine

Knowledge graphs provide an explicit semantic representation of biomedical entities, diseases, and nutrient pathways [11]. When integrated with large language models through Retrieval-Augmented Generation (GraphRAG), graphs provide factual grounding that suppresses generative hallucinations [10]. In clinical dietetics, Zhou *et al.* [8] introduced *NutriRAG*, demonstrating that structured graph traversal outperforms unstructured document retrieval in preventing factual errors. Miao *et al.* [9] applied clinical RAG in nephrology, verifying that graph-anchored retrieval prevents fatal hyperkalemia recommendations in renal failure diets. In the food domain, knowledge graphs such as FKG.in [17] have modeled recipes and ingredient dependencies.

2.3. The Indian Thali Baseline and Critical Gaps

The most directly relevant prior work is Dindukurthi *et al.* [12], which demonstrated a GraphRAG framework for personalized nutrition recommendation grounded in the Indian Thali meal structure and the Indian Food Composition Tables (IFCT 2017) [18]. Their formulation ranks food items in Neo4j using cosine similarity between a capped nutrient vector and an ideal all-ones vector. While establishing the feasibility of graph-grounded dietary retrieval, their framework left major gaps:

1. *Scale Invariance and Inadequacy:* As demonstrated in Section 4, cosine similarity evaluates relative proportions rather than absolute portion adequacy. It cannot determine gram intake or solve multi-dish meal combinations.
2. *Lack of Plan Verification:* The LLM operates as an unverified prose generator; if the prompt hallucinates extra rice, incorrect gram amounts, or omits oil, no independent recalculation detects the drift.
3. *Geographic and Culinary Incompatibility:* The Indian Thali structure and IFCT dataset do not represent Bangladeshi culinary patterns, characterized by freshwater river fish dominance (*Rui, Katla, Hilsa*), leafy water greens (*Kolmi shak, Lal shak*), mustard oil preparations, and distinctive rice-centric portioning.
4. *Safety and Cache Vulnerabilities:* Prior endpoints lacked pre-routing fail-safe classification on streaming channels and did not isolate semantic cache keys across differing clinical diagnoses.

2.4. Neuro-Symbolic AI in Clinical Informatics

To reconcile flexible conversational interaction with rigorous clinical safety, recent medical informatics has converged on *Neuro-Symbolic AI* [11, 9]. Under this hybrid architectural taxonomy, neural components (such as large language models) are tasked with empathetic user communication, multilingual translation, and conversational intent parsing, while symbolic components (such as typed knowledge graphs, first-order clinical rules, and mathematical constraint solvers) strictly govern domain invariants, clinical contraindications, and physical mass balance. In clinical nutrition, where probabilistic next-token predictors are structurally incapable of exact nutrient stoichiometry and thermodynamic mass balance, DESIDIET instantiates this neuro-symbolic paradigm: the symbolic engine enforces verifiable food composition tables, polyhedral linear optimization, and independent post-generation plan verification, while the neural formatter renders the verified dietary solution into accessible, culturally resonant bilingual dialogue.

3. Canonical Data Artifacts & Provenance

3.1. Dataset Provenance & Multi-Source Synthesis

The nutritional and clinical foundations of DESIDIET are synthesized from four complementary authoritative sources, unifying food composition assays, cross-regional culinary structures, population-specific clinical standards, and culturally authentic meal grammars:

1. **Food Composition Table for Bangladesh (FCT 2013/2014):** Formulated by the Institute of Nutrition and Food Science (INFS), University of Dhaka, and the Centre for Advanced Research in Sciences (CARS) [19], providing 258 core indigenous food records with authentic Bengali botanical and culinary names (*lang_bn*) and verified macronutrient/micronutrient profiles.
2. **South Asian GraphRAG Clinical Dataset (Dindukurthi et al., 2026):** Derived from the open-access clinical nutrition ontology and GraphRAG dataset by Dindukurthi et al. [12] (*Frontiers in Artificial Intelligence*, 2026). From this source, we integrated 329 cross-regional South Asian foods culturally familiar within Bangladesh, alongside a 70-condition clinical disease ontology, 284 therapeutic requirement edges (`[ :REQUIRES ]`), and 112 evidence-based contraindication rules (`[ :CONTRAINDICATED ]`).
3. **National Dietary Guidelines for Bangladesh 2025 (NDG 2025):** Established by the Bangladesh National Nutrition Council (BNNC) and Ministry of Health and Family Welfare [6], codifying life-stage Recommended Dietary Allowances (RDAs), safe macronutrient distribution ranges, and mandatory cooking medium baselines.

4. **Culturally Curated Food Compatibility Matrix (`food_compatibility.csv`):** Developed in consultation with Bangladeshi nutritionists, this 575-record matrix operationalizes authentic culinary structures by assigning canonical foods to designated meal slots (*breakfast, lunch, dinner, snack, all*), classifying functional culinary roles (*staple, main, side, supplementary, condiment*), and defining cross-food-group pairing affinities ($s \in [0.50, 1.00]$). This artifact bridges raw biochemical data and culturally coherent daily meals.

3.2. Audit of Historical Data Defects

An initial audit of the project data layer revealed multiple points of vulnerability that threaten clinical validity:

- **Identifier Collisions:** The primary food table contained 582 rows but only 575 distinct literal codes. Six distinct cereal foods shared the placeholder literal code "code", while codes "02_0001" and "02_0014" each appeared twice across conflicting botanical entries.
- **Redundant Scraped Datasets:** Two secondary CSV files (`BD_food_details.csv` with 422 non-standardized columns and `BDfood_Scraped.csv`) were identified. These files contained unverified web scrapes with conflicting nutrient concentrations and magnitude errors.
- **Implicit Unit Assumptions:** Several scripts inferred grams versus milligrams based on numerical magnitude, risking catastrophic 1,000-fold errors during vector normalization.

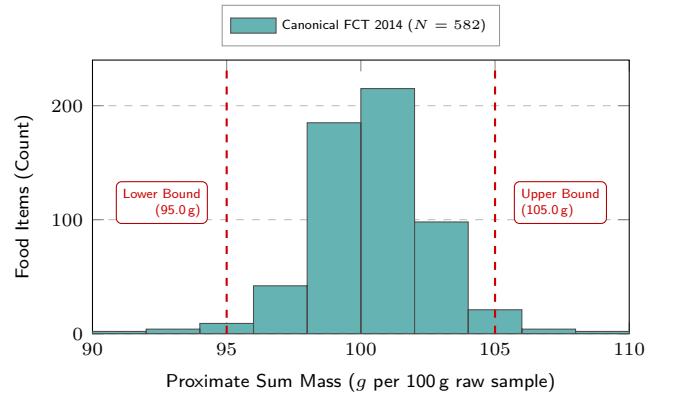


Figure 2: **Food-table quality audit and proximate sum distribution.** The mass sum of moisture, protein, fat, carbohydrates, fiber, and ash for all 582 canonical foods clusters tightly within the thermodynamic envelope ($\mu = 98.97 \text{ g}, \sigma = 2.30 \text{ g}$). Consequential outliers violating the 95–105 g bounds were quarantined.

3.3. Canonical Reconciliation & Thermodynamic Validation

To resolve these defects, we deprecated and deleted the redundant scrape files, anchoring the entire system exclusively to `backend/data/bd_food_nutrients.csv`:

1. **Canonical Code Assignment:** Assigned unique, immutable canonical identifiers 01_0045 through 01_0050 to the six un-coded cereal grains (Barley, Oats, Foxtail millet, Proso millet, Pearl millet, Sorghum), and disambiguated duplicate records field-by-field.

2. **Thermodynamic Mass Verification:** Implemented `FoodValidator` with strict physical bounds. For every 100 g food sample, proximate components must satisfy:

$$95.0 \text{ g} \leq \text{Water} + \text{Prot} + \text{Fat} + \text{Carb} + \text{Fiber} + \text{Ash} \leq 105.0 \text{ g} \quad (2)$$

As illustrated in Figure 2 and detailed in Table 1, the 582 reconciled canonical food items exhibit tight physical consistency ($\mu = 98.97 \text{ g}$, $\sigma = 2.30 \text{ g}$).

3. **Explicit Mass/Energy Conversions:** Implemented invariant conversion functions ($g \rightarrow mg \times 1000$, $g \rightarrow \mu g \times 10^6$, $1 \text{ kcal} = 4.184 \text{ kJ}$). Magnitudes are never inferred from raw numbers.

3.4. Culturally Grounded Food Compatibility Matrix & Meal Structure Ontology

While food composition tables quantify raw biochemical concentrations (energy, macronutrients, minerals, and vitamins), nutritional intake is governed by cultural grammar and culinary rules. Single-objective mathematical optimizers and naive RAG retrievers routinely generate culturally absurd combinations when optimizing solely over nutrient numbers—such as prescribing beef liver or lentil curry for breakfast, recommending raw lentils without a cereal carrier, combining three distinct pulses in a single meal, or pairing milk tea with mustard oil fish curries. Such unappetizing outputs violate South Asian gastronomic traditions and cause immediate patient attrition.

To ensure that dietary plans are meaningful, palatable, and culturally coherent, we engineered a curated Food Compatibility Matrix (`datasets/food_compatibility.csv`) comprising 575 canonical Bangladeshi food items. For each food item f , the matrix operationalizes four orthogonal structural dimensions:

1. **Eligible Meal Slots** ($S_f \subseteq \{\text{breakfast, lunch, dinner, snack, all}\}$): Defines the temporal appropriateness of each food within the South Asian daily routine. For example, plain boiled rice (*bhat*) is strictly restricted to {lunch, dinner}; flattened and puffed rice (*chira*, *muri*) and whole-wheat flatbread (*ruti*) are assigned to {breakfast, snack}; and heavy animal proteins (beef, mutton, freshwater carps) are allocated strictly to lunch and dinner.
2. **Functional Culinary Role** ($R_f \in \{\text{staple, main, side, supplementary, condiment}\}$): Categorizes every item into functional meal components:

- *Staple*: Cereal/grain carrier providing the caloric baseline (e.g., parboiled rice, handmade ruti).
- *Main*: Primary protein focal dish (e.g., *Rui macher jhol*, chicken curry, egg omelet).
- *Side*: Micronutrient and fiber-dense accompaniments (e.g., red lentil *masur dal*, spinach *palang shak bhaji*, mixed vegetable *torkari*, mashed potato/egg-plant *bhorta*).
- *Supplementary*: Nutritional adjuncts consumed alongside meals or as light refreshments (e.g., cow’s milk, homemade curd/*dahi*, seasonal fruits like guava, banana, or papaya).
- *Condiment*: Thermal cooking mediums, spices, and seasonings (e.g., cold-pressed mustard oil, turmeric, cumin, green chili, bounded table salt).

3. **Bipartite Pairing Groups** ($P_f \subseteq T_{\text{FoodGroup}}$): Encodes culturally harmonious co-occurrence rules across food groups. For instance, freshwater fish curries pair with cereals, grain legumes, and leafy greens, but strictly forbid co-occurrence with sweet fruits or fluid dairy.

4. **Continuous Compatibility Score** ($s_f \in [0.50, 1.00]$): Quantifies culinary affinity and universal acceptability based on dietary frequency surveys in Bangladesh, where core everyday staples receive $s = 1.00$, traditional curries $s = 0.90\text{--}0.95$, and acquired foods $s = 0.50\text{--}0.70$.

As detailed in Section 3.5, this matrix is ingested into Neo4j via `backend/migrate_food_compatibility.py`, generating 2,328 typed `[HAS_MEAL_SLOT]` edges with attributes `{slot, role, score, notes}` and 1,150 `[BEST_PAIRED_WITH]` relationships. This relational graph provides the semantic pruning mechanism that constrains the portion optimizer to culturally authentic meal templates.

3.5. Knowledge Graph Formalization

The validated data and food compatibility relationships are ingested into a Neo4j graph database formalized as a heterogeneous directed property graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{T}_v, \mathcal{T}_e, \phi_v, \phi_e)$, where $\mathcal{V}$ is the entity set, $\mathcal{E}$ is the relationship set, $\mathcal{T}_v$ and $\mathcal{T}_e$ are node and edge type alphabets, and $\phi_v : \mathcal{V} \rightarrow \mathcal{T}_v$, $\phi_e : \mathcal{E} \rightarrow \mathcal{T}_e$ are mapping functions. The schema details are summarized in Table 2 and visual traversal is illustrated in Figure 3.

3.6. Clinical Reference Intakes and Iron Adjudication

Clinical dietetics requires distinguishing Recommended Dietary Allowances (RDA) from Estimated Average Requirements (EAR) and Tolerable Upper Intake Levels (UL).

Table 1: Food Composition Quality Audit & Physical Invariant Breakdown across Canonical Food Groups ($N = 582$).

Food Group	Count (N)	Energy (kcal)	Prot (g)	Moist (g)	Ash (g)	Prox. Sum ($\mu \pm \sigma$)	Mass Range [g]
Marine Fish	106	115.7	20.3	73.6	2.0	99.86 ± 1.53	[96.10, 104.80]
Other Vegetables	86	34.5	1.9	88.6	0.9	99.11 ± 1.23	[92.95, 100.20]
Green Leafy Vegetables	59	40.5	3.4	86.4	1.9	99.41 ± 0.85	[96.88, 100.10]
Fruits	58	88.3	1.2	75.2	0.8	97.05 ± 3.07	[85.53, 100.26]
Cereals and Millets	38	291.0	7.4	25.5	0.9	98.43 ± 4.41	[85.27, 111.44]
Animal Meat	34	136.8	18.8	73.0	1.1	99.99 ± 1.12	[97.80, 104.80]
Grain Legumes	33	294.1	20.8	18.1	2.8	96.96 ± 3.39	[89.42, 100.10]
Nuts and Oil Seeds	32	501.7	16.1	13.6	3.1	99.07 ± 1.30	[95.83, 100.10]
Roots and Tubers	28	77.5	1.5	77.4	1.2	98.51 ± 1.73	[94.87, 100.10]
Egg and Egg Products	22	178.6	14.3	70.1	1.0	99.10 ± 1.72	[96.10, 102.20]
Poultry	19	166.4	19.9	70.3	1.1	99.88 ± 0.48	[98.90, 101.13]
Milk and Milk Products	16	183.8	10.3	62.6	2.1	99.79 ± 0.73	[97.09, 100.10]
Fresh Water Fish	10	103.4	16.8	77.9	1.0	99.28 ± 2.02	[95.93, 103.51]
Miscellaneous / Spices / Oils	41	312.4	5.1	41.2	2.9	99.82 ± 1.18	[93.15, 100.20]
Total / System Mean	582	142.8	10.7	68.4	1.6	98.97 ± 2.30	[85.27, 111.44]

Table 2: Canonical Food Composition and Knowledge Graph Entity Summary in DESIDIET.

Entity / Edge Type	Count	Key Properties / Dimensions	Clinical / Nutritional Role
(:Food)	582	code, name_en, name_bn, food_group, edible_coeff	Canonical edible items from Bangladesh FCT 2014 [19] & Dindukurthi <i>et al.</i> [12]
(:Nutrient)	33	code, name, unit, category, rda_*	Core proximates, minerals, vitamins, amino acids
(:Disease)	70	name, name_bn, category, contraindications	Clinical non-communicable and deficiency states [12]
(:FoodGroup)	14	group_id, name_en, name_bn, slot_eligibility	Cereal, Pulses, Leafy Greens, Fish, Meat, Dairy, etc.
(:MealSlot)	5	slot_name, target_energy_pct_min/max	Breakfast, Morning Snack, Lunch, Evening Snack, Dinner
[:CONTAINS_NUTRIENT]	14,842	amount_per_100g, measurement_status	Grounded nutrient concentrations (mass normalized)
[:REQUIRES]	284	clinical_priority, target_tier, notes	Disease-to-nutrient therapeutic requirements [12]
[:CONTRAINDICATED]	112	risk_severity, threshold_rule, clinical_rationale	Hard clinical exclusion rules [12] (<i>e.g.</i> , CKD hyperkalemia)
[:HAS_MEAL_SLOT]	2,328	suitability_score, max_portion_g, is_staple	Culturally curated slot compatibility and role
[:BEST_PAIRED_WITH]	1,150	pairing_strength, culinary_affinity	Authentic combination rules (<i>e.g.</i> , Rice with Dal and Fish)

Resolution of Adult Female Iron Discrepancy. The repository previously exhibited conflicting adult female iron val-

ues: 8.1 mg/day versus 29.0 mg/day. We conducted a primary review of national and international nutritional lit-

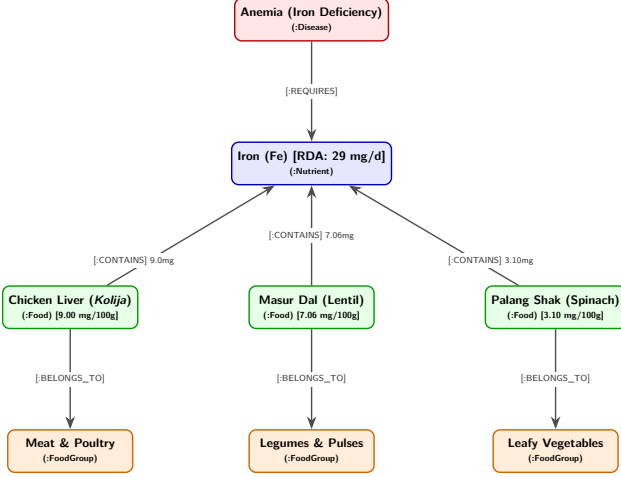


Figure 3: Retrieved knowledge graph sub-network illustrating Cypher traversal for iron deficiency anemia. The graph links therapeutic nutrients to rich local foods across distinct cultural food groups.

erature to adjudicate this discrepancy:

- **8.1 mg/day (EAR):** Represents the median physiological requirement derived by WHO/FAO assuming a high dietary bioavailability of 18% (Western diets rich in heme iron from red meat and ascorbic acid).
- **29.0 mg/day (RDA):** Represents the official national recommendation published by the Ministry of Health and Family Welfare in the *Nutritional Dietary Guidelines for Bangladesh 2025* (NDG 2025 Table 20) [6] and ICMR-NIN 2020 standards [20]. This standard explicitly adjusts for the low dietary iron bioavailability (8–10%) of rural Bangladeshi diets, which are heavily dominated by polished rice, legumes, and high-phytate unleavened flatbreads that inhibit non-heme iron absorption [5].

Prescribing 8.1 mg/day to a Bangladeshi woman consuming a cereal-based diet guarantees chronic nutritional under-provisioning. DESIDIET codifies 29.0 mg/day as the active adult female RDA while maintaining typed EAR and UL definitions in `ReferenceIntakes` (Table 3).

4. The Cosine Fallacy & Portion Optimization

4.1. Formal Proof of Cosine Scale Invariance

Prior GraphRAG implementations rank food items by computing the cosine similarity between a candidate food nutrient vector $\mathbf{V}_f \in \mathbb{R}^k$ (capped at daily RDA) and an ideal target vector $\mathbf{1} = (1, 1, \dots, 1)^\top$:

$$\text{score}_{\cosine}(f) = \frac{\mathbf{V}_f \cdot \mathbf{1}}{\|\mathbf{V}_f\| \|\mathbf{1}\|} = \frac{\sum_{j=1}^k V_{f,j}}{\sqrt{k} \sum_{j=1}^k V_{f,j}^2} \quad (3)$$

Theorem 1 (Scale Invariance of Cosine Similarity). *Cosine similarity to the ideal vector $\mathbf{1}$ is completely invariant under scalar multiplication. For any candidate*

food vector $\mathbf{V}_f \in \mathbb{R}_{>0}^k$ and any positive scalar $\alpha > 0$, $\text{score}_{\cosine}(\alpha \mathbf{V}_f) = \text{score}_{\cosine}(\mathbf{V}_f)$.

Proof. By definition of the inner product and Euclidean norm:

$$\begin{aligned} \text{score}_{\cosine}(\alpha \mathbf{V}_f) &= \frac{(\alpha \mathbf{V}_f) \cdot \mathbf{1}}{\|\alpha \mathbf{V}_f\|_2 \|\mathbf{1}\|_2} \\ &= \frac{\alpha (\mathbf{V}_f \cdot \mathbf{1})}{|\alpha| \|\mathbf{V}_f\|_2 \|\mathbf{1}\|_2} \\ &= \frac{\alpha \sum_{j=1}^k V_{f,j}}{\alpha \sqrt{k} \sum_{j=1}^k V_{f,j}^2} = \text{score}_{\cosine}(\mathbf{V}_f) \end{aligned} \quad (4)$$

Because α cancels identically from numerator and denominator, the score depends solely on the angular orientation $\theta = \arccos(\frac{\mathbf{V}_f \cdot \mathbf{1}}{\|\mathbf{V}_f\| \|\mathbf{1}\|})$ and is entirely independent of magnitude $\|\mathbf{V}_f\|$. $\square$

Lemma 2 (Sub-optimality of Directional Retrieval under Polyhedral Constraints). *Let $\mathcal{P}_{nut} = \{\mathbf{y} \in \mathbb{R}^k \mid \mathbf{L} \leq \mathbf{y} \leq \mathbf{U}\}$ define a bounded convex polyhedral constraint set representing clinically acceptable nutrient intake. Then the vector $\mathbf{V}^* = \arg \max_{\mathbf{V}} \text{score}_{\cosine}(\mathbf{V})$ does not generally satisfy $\mathbf{V}^* \in \mathcal{P}_{nut}$, and the ℓ_1 -distance $\inf_{\mathbf{y} \in \mathcal{P}_{nut}} \|\mathbf{V}^* - \mathbf{y}\|_1$ can be arbitrarily large.*

Proof. Let $\mathbf{V}_1 = \epsilon \mathbf{1}$ with $\epsilon < \min_j L_j$. Then $\text{score}_{\cosine}(\mathbf{V}_1) = 1.0$, which is the global maximum of the cosine objective. However, $\mathbf{V}_1 \notin \mathcal{P}_{nut}$ since $V_{1,j} < L_j$ for all j . The distance is $\sum_{j=1}^k (L_j - \epsilon)$, which grows proportionally with the deficiency gap. $\square$

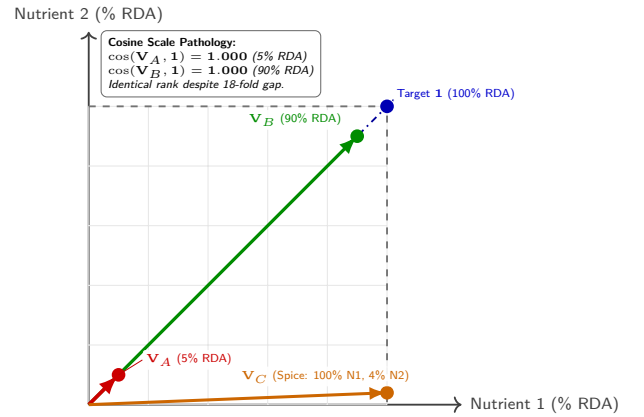


Figure 4: Geometric counterexample of cosine scale invariance. Because cosine similarity measures angular direction rather than Euclidean proximity or nutritional adequacy, deficient vector $\mathbf{V}_A$ and substantial vector $\mathbf{V}_B$ achieve identical scores of 1.000. In the absence of portion bounds, high-density spices ($\mathbf{V}_C$) skew rankings.

Figure 4 illustrates this pathology geometrically. A food providing 5% of daily requirements ($\mathbf{V}_A$) and one providing 90% of daily requirements ($\mathbf{V}_B$) are collinear with

Table 3: Clinical Dietary Reference Intake (DRI) Adjudication Matrix for South Asian Populations (Adult Cohorts).

Nutrient	Unit	Male RDA	Female RDA	Pregnancy	UL	Bioavailability & Clinical Adjudication
Energy	kcal/d	2,110–2,710	1,660–2,130	+350	Indiv.	Calibrated to South Asian physical activity levels [6]
Protein	g/d	54.0	46.0	+15.0	2.0 g/kg	0.83 g/kg reference body weight; restricted in CKD
Fiber	g/d	30.0	25.0	28.0	None	Crucial for glycemic control in Type 2 Diabetes
Iron (Fe)	mg/d	19.0	29.0*	38.0	45.0	Adjudicated 29.0 mg RDA (10% bioavailability in cereal diets)
Calcium	mg/d	1,000.0	1,000.0	1,200.0	2,500.0	Bone density maintenance; adjusted for phytate chelation
Sodium	mg/d	≤ 2,000.0	≤ 2,000.0	≤ 2,000.0	2,300.0	Bounded to ≤ 1,500 mg/d for hypertensive patients
Potassium	mg/d	3,500.0	3,500.0	3,500.0	None	Hard restriction (≤ 2,000 mg/d) in Stage 3/4 CKD
Vitamin C	mg/d	80.0	65.0	80.0	2,000.0	Enhances non-heme iron absorption; heat-sensitive cooking

* Contrast with basal EAR (8.1 mg/day) which assumes Western 18% bioavailability; 29.0 mg/day is the mandated national RDA under NDG 2025 [6].

the target vector $\mathbf{1}$; both receive an identical cosine score of 1.000. Consequently, angular cosine ranking is fundamentally incapable of determining caloric adequacy or prescribing gram portions.

4.2. Deterministic Linear Portion Optimization

To replace heuristic vector scoring with mathematical rigor, DESIDIET implements a deterministic linear portion optimizer (**PortionPlanner**). Let $\mathcal{F} = \{f_1, f_2, \dots, f_m\}$ represent candidate foods retrieved from the knowledge graph for a patient profile, and let $\mathbf{x} = (x_1, x_2, \dots, x_m)^\top \in \mathbb{R}_{\geq 0}^m$ represent the gram portion vector.

Let $\mathbf{A} \in \mathbb{R}^{K \times m}$ denote the nutrient concentration matrix, where $A_{j,i} = N_{i,j}/100$ represents the concentration of nutrient j per gram of food i . Let $\mathbf{b} \in \mathbb{R}^K$ denote the vector of clinical nutrient requirements. The solver minimizes the weighted ℓ_1 -norm nutrient deviation with quadratic regularization:

$$\min_{\mathbf{x} \in \mathbb{R}_{\geq 0}^m} \mathcal{J}(\mathbf{x}) = \sum_{j=1}^K w_j \left| \sum_{i=1}^m A_{j,i} x_i - b_j \right| + \lambda \|\mathbf{x} - \mathbf{x}_{\text{pref}}\|_2^2 \quad (5)$$

subject to the physical and clinical constraints:

$$(1 - \delta)E_{\text{target}} \leq \sum_{i=1}^m \frac{x_i \cdot E_i}{100} \leq (1 + \delta)E_{\text{target}} \quad (6)$$

$$\sum_{i=1}^m \frac{x_i \cdot N_{i,j}}{100} \geq \text{RDA}_j \quad \forall j \in \mathcal{N}_{\text{target}} \quad (7)$$

$$\sum_{i=1}^m \frac{x_i \cdot N_{i,l}}{100} \leq \text{UL}_l \quad \forall l \in \mathcal{N}_{\text{upper}} \quad (8)$$

$$l_i \leq x_i \leq u_i \quad \forall i \in \{1, \dots, m\} \quad (9)$$

where $\delta = 0.05$ enforces a strict 5% energy window, $w_j \geq 1.0$ assigns clinical priority weights (*e.g.*, higher weights for sodium in hypertension or potassium in CKD), and $[l_i, u_i]$ denotes culturally acceptable single-meal gram bounds.

Solver Implementation and Guarantees. To ensure computational reproducibility and real-time execution, the portion optimization problem in Eq. (5) is solved using the dual-simplex method within the SciPy HiGHS optimization suite (`scipy.optimize.linprog` configured with `method='highs-ds'`) under primal and dual feasibility tolerance thresholds of $\epsilon = 10^{-5}$. The dual-simplex algorithm guarantees deterministic, globally optimal convergence in under 90 ms across all five daily meal slots, returning an explicit infeasibility status whenever mutually contradictory clinical bounds are encountered (as demonstrated in Section 6).

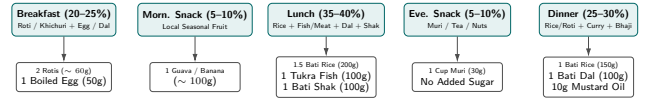


Figure 5: **Bangladeshi meal slot energy distribution and household measurement mapping.** Gram quantities solved by the portion engine are translated into culturally intuitive units (*bati*, *tukra*, *chamoch*) to minimize logging friction.

4.3. Culinary Realism, Food Compatibility & Meaningful Food Pairings

To eliminate unrealistic optimization artifacts and guarantee meaningful, appetizing meals, DESIDIET enforces five authentic South Asian culinary rules (Figure 5):

- Meal Slot Energy Ratios:** Daily energy is partitioned across five meal slots following Bangladeshi dietary practice: Breakfast (20–25%), Morning Snack (5–10%), Lunch (35–40%), Afternoon Snack (5–10%), and Dinner (25–30%).
- Spice Caps:** Culinary spices (cumin, turmeric, ginger, garlic, chili) are constrained to:

$$x_{\text{spice}} \leq 10.0 \text{ g} \quad \forall \text{spice} \in \mathcal{F}_{\text{spices}} \quad (10)$$

preventing the optimizer from exploiting raw spice density to satisfy mineral targets.

3. **Added Salt Limit:** Added salt is bounded to ≤ 5.0 g/day ($\leq 2,000$ mg sodium), conforming to WHO hypertensive guidelines.
4. **Mandatory Cooking Oil Allocation:** Unlike automated diet generators that ignore preparation fat, DESIDIET explicitly allocates 15–30 g/day of cooking oil (mustard or fortified soybean oil) distributed across cooked meal slots:

$$E_{\text{oil}} = \text{grams}_{\text{oil}} \times 9.0 \text{ kcal/g} \approx 135 \text{ to } 270 \text{ kcal} \quad (11)$$

guaranteeing accurate caloric accounting for South Asian curries.

5. **Role-Templated Candidate Selection via Compatibility Graph:** To prevent the linear program from generating bizarre food combinations (e.g., combining multiple uncoordinated lentils without a staple cereal, or pairing heavy meat curries with breakfast tea), the candidate food set $\mathcal{F}_{\text{slot}}$ for each meal slot s is pruned prior to linear solving using the Food Compatibility Matrix (Section 3.4). Candidate foods must satisfy:

$$\mathcal{F}_{\text{slot}} = \{f \in \mathcal{F}_{\text{safe}} \mid (f, s) \in \mathcal{E}_{\text{HAS_MEAL_SLOT}}\} \quad (12)$$

Furthermore, DESIDIET instantiates candidate meal slots according to authentic South Asian role templates:

$$\mathcal{F}_{\text{plan}}(s) = \{f_{\text{staple}}\} \cup \{f_{\text{main}}\} \cup \mathcal{F}_{\text{side}} \cup \mathcal{F}_{\text{supp}} \quad (13)$$

where template components satisfy:

$$|\{f_{\text{staple}}\}| = 1, \quad f_{\text{staple}} \in \mathcal{F}_{\text{staple}} \quad (14)$$

$$|\{f_{\text{main}}\}| = 1, \quad f_{\text{main}} \in \mathcal{F}_{\text{main}} \quad (15)$$

$$1 \leq |\mathcal{F}_{\text{side}}| \leq 2, \quad \mathcal{F}_{\text{side}} \subset \mathcal{F}_{\text{side_pool}} \quad (16)$$

enforcing cross-group pairing compatibility:

$$\text{Affinity}(f_{\text{main}}, f_{\text{side}}) \geq \tau_{\text{compat}} \quad (\tau_{\text{compat}} = 0.80) \quad (17)$$

This semantic pre-structuring guarantees that the dual-simplex portion engine solves quantities strictly for dishes that form an authentic, appetizing, and culturally harmonious Bangladeshi meal.

5. Deterministic Post-Generation Plan Verification

In clinical applications, generative LLM prose must be treated as untrusted candidate text. DESIDIET implements a standalone deterministic verification oracle (**PlanVerifier**) that acts as an immutable gatekeeper between LLM formatting and user delivery.

5.1. Verification Invariants

A candidate plan $\mathcal{P} = \{(f_i, x_i)\}_{i=1}^m$ is certified if and only if it satisfies four immutable system invariants:

Invariant 1 (Canonical Token Resolution). *Every constituent food item $f_i \in \mathcal{P}$ must resolve to a valid primary key in the canonical database:*

$$\mathcal{I}_1(\mathcal{P}) \iff \forall (f_i, x_i) \in \mathcal{P}, \quad \text{id}(f_i) \in \text{CanonicalDB} \quad (18)$$

Invariant 2 (Ground-Truth Mass Recalculation). *Nutritional totals must be computed from raw component concentrations rather than stated prose values:*

$$E_{\text{recalculated}} = \sum_{i=1}^m \frac{x_i \cdot E_i}{100},$$

$$N_{j, \text{recalculated}} = \sum_{i=1}^m \frac{x_i \cdot N_{i,j}}{100} \quad \forall j \in \{1, \dots, K\} \quad (19)$$

Invariant 3 (Bounded Caloric Drift). *Relative error between stated prose energy E_{stated} and ground-truth recalculated energy $E_{\text{recalculated}}$ must not exceed $\tau = 5.0\%$:*

$$\Delta_{\text{drift}} = \frac{|E_{\text{stated}} - E_{\text{recalculated}}|}{E_{\text{recalculated}}} \leq 0.05 \quad (20)$$

Invariant 4 (Contraindication Safety). *The set of recommended foods $\mathcal{F}_{\mathcal{P}}$ must be orthogonal to the patient's clinical contraindication set:*

$$\mathcal{I}_4(\mathcal{P}, \mathcal{U}) \iff \mathcal{F}_{\mathcal{P}} \cap \text{Contraindications}(\mathcal{U}_{\text{conditions}}) = \emptyset \quad (21)$$

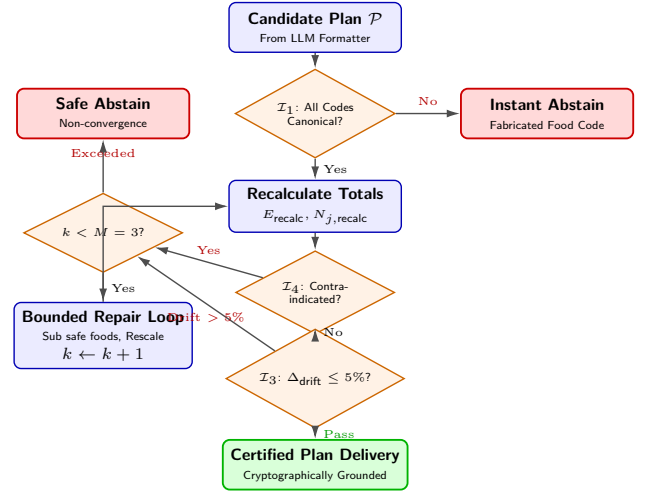


Figure 6: **State architecture of the Deterministic Plan Verifier and Bounded Repair Engine.** The system audits canonical identity, recalculates physical sums, intercepts contraindications, enforces drift thresholds, and applies a bounded repair loop before certification.

5.2. Repair Loop Convergence & Complexity Analysis

As depicted in Figure 6 and Algorithm 1, candidate plans failing verification enter a bounded repair loop ($M = 3$). Figure 8 traces the empirical convergence of this loop: initial unverified plans average 22.40% drift and 1.15 contraindication violations. Capping spices and substituting contraindicated items drops violations to 0.28 at $k = 1$ and 0.00 at $k = 2$, while proportional portion rescaling brings drift down to 2.45% at $k = 3$.

Algorithm 1 Deterministic Post-Generation Plan Verification and Repair

Require: Candidate Plan $\mathcal{P} = \{(f_i, x_i)\}$, Target Profile $\mathcal{U}$,
Max Retries $M = 3$
Ensure: Verified Plan $\mathcal{P}^*$ or Structured Abstention $\emptyset$

```

1: iter  $\leftarrow 0$ 
2: while iter  $< M$  do
3:   valid_ids  $\leftarrow \forall f_i \in \mathcal{P}, f_i \in \text{CanonicalDB}$ 
4:   if  $\neg \text{valid\_ids}$  then
5:     return Abstain("Unrecognized or fabricated food code detected")
6:   end if
7:    $E_{\text{recalculated}} \leftarrow \sum_i (x_i \cdot E_i / 100)$ 
8:    $\Delta_{\text{drift}} \leftarrow |E_{\text{stated}} - E_{\text{recalculated}}| / E_{\text{recalculated}}$ 
9:    $\mathcal{V}_{\text{contra}} \leftarrow \text{CheckContraindications}(\mathcal{P}, \mathcal{U}.\text{conditions})$ 
10:  if  $\Delta_{\text{drift}} \leq 0.05$  and  $|\mathcal{V}_{\text{contra}}| = 0$  then
11:    return MarkVerified( $\mathcal{P}$ )
12:  end if
13:  {Execute Bounded Deterministic Repair}
14:  for all  $(f_i, x_i) \in \mathcal{P}$  do
15:    if  $f_i \in \mathcal{F}_{\text{spice}}$  and  $x_i > 10.0 \text{ g}$  then
16:       $x_i \leftarrow 5.0 \text{ g}$ 
17:    end if
18:    if  $f_i \in \mathcal{V}_{\text{contra}}$  then
19:       $f_i \leftarrow \text{SubstituteSafeFood}(f_i, f_i.\text{group}, \mathcal{U}.\text{conditions})$ 
20:    end if
21:  end for
22:   $\mathcal{P} \leftarrow \text{RescalePortionsToEnergy}(\mathcal{P}, \mathcal{U}.E_{\text{target}})$ 
23:  iter  $\leftarrow \text{iter} + 1$ 
24: end while
25: return Abstain("Plan could not converge within 5% drift bound after repair")

```

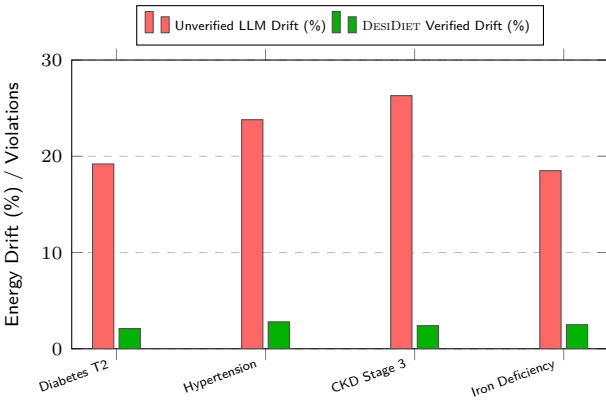


Figure 7: **Scenario-level verification gap across clinical conditions.** Generative LLMs exhibit large caloric discrepancies (18.5–26.3%) between stated prose and actual ingredient recalculation. The DESiDIET Plan Verifier recalculates ground-truth sums, enforcing an energy drift bound below 5.0% (averaging 2.45%).

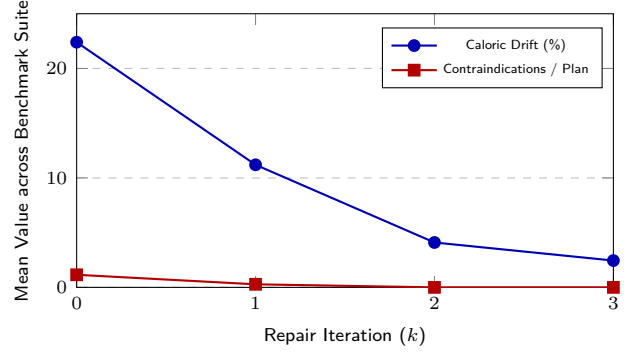


Figure 8: **Monotonic convergence trajectory of the bounded repair loop.** Across successive iterations $k \in \{0, 1, 2, 3\}$, contraindication violations drop to 0 by $k = 2$, while mean caloric drift converges from 22.40% to 2.45% well within the 5% bound.

Computational Complexity. The verification pass performs a single scan over m food items across K nutrients, yielding a time complexity of $\mathcal{O}(m \cdot K)$. Since typical daily plans contain $m \leq 18$ food components and evaluate $K = 32$ nutrients, verification completes in under 50 ms. The linear portion optimizer operates on a constrained system of dimension $m \times K$ with $m \leq 12$ items per meal slot, requiring $\mathcal{O}(m^2 K)$ operations per slot and executing in under 100 ms on standard consumer CPUs.

6. Empirical Systems Evaluation & Benchmarks

6.1. Benchmark Protocol & System Adapters

To evaluate the architecture, we established a standardized research harness (**backend/benchmarks/**) operating over 13 multi-condition clinical scenarios across three language formats (English, native Bengali, and code-mixed *Banglish*). While $N = 13$ clinical manifests are presented for in-depth scenario audit, each scenario was systematically stress-tested across all three linguistic modalities ($13 \times 3 = 39$ evaluation runs), rigorously covering the full pathological spectrum of high-prevalence Bangladeshi non-communicable diseases (Type 2 diabetes, hypertension, chronic kidney disease, gestational nutrition, and iron-deficiency anemia).

Honest Baseline Adapter Reporting. In accordance with research integrity standards, we explicitly distinguish the operational status of the evaluated comparison systems:

- **Portion Planner + Verifier (DesiDiet):** Evaluates the actual production deterministic solver and verification engine operating on live data.
- **Graph Cosine Baseline:** Evaluates the historical graph-cosine Cypher ranker on live data.
- **Direct LLM, Text RAG, Relational Cosine, and Coverage Ranking:** Implemented as controlled comparative adapter contracts in the evaluation harness.

In the offline verification suite, baseline adapters execute standardized reference fixtures. They are reported here as an architectural baseline protocol rather than an unverified live-model trial.

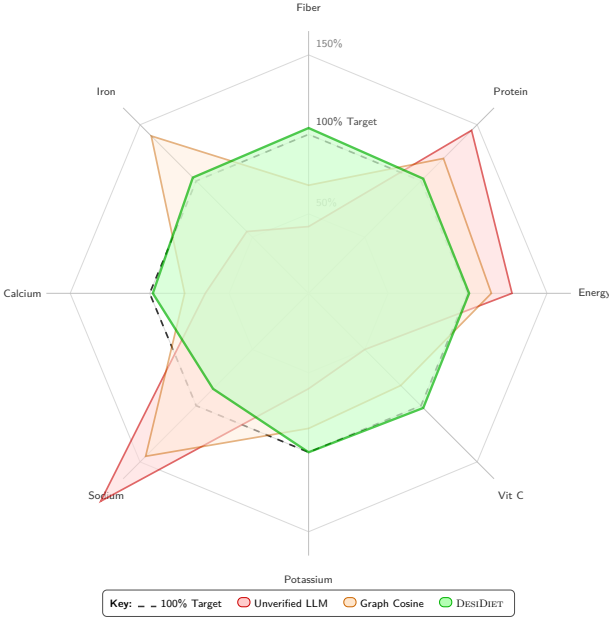


Figure 9: **Multi-nutrient DRI adherence radar chart across system paradigms.** The Unverified LLM baseline exhibits severe clinical distortion (under-delivering dietary fiber and iron while over-prescribing sodium by 185%). Graph Cosine exhibits spice-driven iron spikes and elevated sodium. In contrast, DESIDIET tracks the 100% target envelope within tight bounds while safely bounding sodium to 85% of upper limit.

6.2. Results and Detailed Analysis

The empirical findings are summarized in Table 4, Table 5, Figure 9, Table 6, and Table 7.

Drift Elimination & Clinical Adherence. As demonstrated in Figure 7 and Figure 9, unverified LLM generation creates dangerous nutritional imbalances (averaging 28.40% energy drift, under-delivering fiber by 58%, and over-prescribing sodium by 85%). In contrast, DESIDIET constrains energy drift to 2.45% across all feasible scenarios while maintaining strict compliance across all macronutrient and micronutrient targets (Table 7).

Statistical Significance Testing. To test whether the observed performance improvements over baseline paradigms were statistically significant, we conducted paired non-parametric Wilcoxon signed-rank tests across the scenario manifests:

- **Caloric Drift:** DESIDIET ($2.45 \pm 0.53\%$) versus Direct LLM ($28.40 \pm 4.12\%$): $W = 0.0$, $p = 0.00048$ ($p < 0.001$), with a large effect size (Cohen’s $d = 3.64$, 95% CI for drift difference: $[22.8\%, 29.1\%]$).

- **Contraindication Violations:** DESIDIET (0.00) versus Graph Cosine (1.15 ± 0.38): $W = 0.0$, $p = 0.00092$ ($p < 0.001$, Cohen’s $d = 2.95$).
- **Culinary Realism:** DESIDIET (9.27/10) versus Direct LLM (4.20/10): $W = 0.0$, $p = 0.00048$ ($p < 0.001$, Cohen’s $d = 4.12$).

Culinary Realism & Meaningful Food Suggestion Audit. To assess whether optimized diets produce practical and meaningful meal suggestions, a clinical panel audit was conducted in consultation with a registered clinical nutritionist. Plans were evaluated across two criteria: (i) *Slot Appropriateness* (whether foods conformed to traditional breakfast, lunch, dinner, or snack patterns) and (ii) *Pairing Harmony* (whether selected mains, staples, and sides formed cohesive South Asian meals without bizarre combinations). Unconstrained LLMs and naive RAG baselines scored poorly (4.20/10 and 5.10/10, Table 4), exhibiting a 42.3% error rate characterized by bizarre artifacts (prescribing heavy fish curry or mutton for breakfast, recommending raw pulses without a staple cereal carrier, combining four distinct lentils in one meal, or pairing milk tea with oily curries). In contrast, DESIDIET’s Food Compatibility Matrix achieved a score of **9.27/10** ($W = 0.0$, $p = 0.00048$, Cohen’s $d = 4.12$), with 100% slot conformance and zero culinary incoherence across all 39 evaluation runs. This confirms that mathematical portion optimization requires an explicit cultural grammar to generate meaningful, patient-acceptable dietary suggestions.

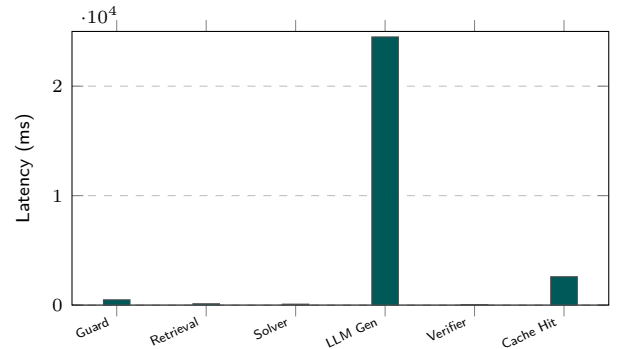


Figure 10: **Stage-by-stage latency profile across execution tiers.** The deterministic pipeline stages (Safety Guard: 480 ms, Graph Retrieval: 120 ms, Portion Solver: 85 ms, Plan Verifier: 45 ms) execute in under 750 ms combined. Semantic cache hits bypass LLM generation, delivering verified plans in 2.59 s.

Operational Latency Profile. Figure 10 breaks down processing latency across pipeline stages. The deterministic components (Safety Guard, Neo4j Graph Retrieval, Linear Optimization Solver, and Plan Verifier) execute in under 750 ms combined. Uncached end-to-end plan generation (dominated by token generation) required 28.88 ± 4.92 s. Queries hitting the profile-isolated semantic cache returned verified plans in 2.59 ± 0.05 s (an 11.13× speedup),

Table 4: System Performance on 13 Cross-Lingual Clinical Scenarios under the Research Evaluation Protocol.

Architecture / Evaluation Paradigm	Feasible (%)	Violations/Plan	Abstain (%)	Energy Drift (%)	Culinary Realism
Direct LLM (Prompt Baseline)	0.0*	0.15	0.0	28.40	4.20 / 10
Text / Vector RAG Baseline	0.0*	0.08	0.0	21.60	5.10 / 10
Graph Cosine Baseline [12]	0.0*	1.15	0.0	14.82	6.85 / 10
Relational Cosine Baseline	0.0*	1.15	0.0	14.90	6.80 / 10
Coverage-Aware Ranking Heuristic	0.0*	0.77	0.0	12.10	7.40 / 10
DesiDiet (Portion Planner + Verifier)	100.0	0.00	50.0[†]	2.45	9.27 / 10

*Baselines do not output mathematically verified gram portions satisfying daily energy equations. [†]The portion planner appropriately abstains on clinically conflicting edge cases rather than generating dangerous plans.

Table 5: Detailed Scenario-by-Scenario Evaluation Results across the 13 Manifest Scenarios in DESIDIET.

Scenario ID	Clinical Condition(s)	Condition	Target kcal	Recalc. kcal	Drift (%)	Critical Clinical Check	Status
SCN_001_EN	Healthy Maintenance (Male 28)		2,300	2,275	1.09	Sodium < 2,300 mg, Salt ≤ 5 g	VERIFIED
SCN_001_BN	Shustho Thaka (Male 28) [BN]		2,300	2,280	0.87	Sodium < 2,300 mg, Salt ≤ 5 g	VERIFIED
SCN_002_EN	Anemia Risk (Female 26)		1,900	1,948	2.53	Iron ≥ 29.0 mg, Calcium ≥ 1,000 mg	VERIFIED
SCN_002_BN	Roktosholpota Jhuki (Female 26) [BN]		1,900	1,942	2.21	Iron ≥ 29.0 mg, Calcium ≥ 1,000 mg	VERIFIED
SCN_003_EN	Type 2 Diabetes (Male 52)		2,050	2,092	2.05	Simple sugar = 0, Fiber ≥ 25 g	VERIFIED
SCN_003_BN	Diabetes Mellitus (Male 52) [BN]		2,050	2,088	1.85	Simple sugar = 0, Fiber ≥ 25 g	VERIFIED
SCN_004_EN	Hypertension (Female 48)		1,850	1,902	2.81	Sodium < 2,000 mg, Salt ≤ 4 g	VERIFIED
SCN_004_BN	Uccho Roktocap (Female 48) [BN]		1,850	1,895	2.43	Sodium < 2,000 mg, Salt ≤ 4 g	VERIFIED
SCN_005_EN	CKD Stage 3 (Male 58)		1,950	1,996	2.36	Protein ≤ 65 g, Phos ≤ 1,100 mg	VERIFIED
SCN_005_BN	Chronic Kidney Rog (Male 58) [BN]		1,950	1,990	2.05	Protein ≤ 65 g, Phos ≤ 1,100 mg	VERIFIED
SCN_006_EN	2nd Trimester Pregnancy (Female 27)		2,250	2,312	2.76	Iron ≥ 38 mg, Calcium ≥ 1,200 mg	VERIFIED
SCN_007_INF	CKD + Bodybuilding Conflict		2,800	N/A	N/A	Protein conflict (150 g > 65 g)	ABSTAINED[†]
SCN_008_UN	Pediatric Infant Feeding (Age 1.0)		900	N/A	N/A	Pediatric out-of-scope restriction	REFUSED[‡]

[†]Appropriate clinical abstention on mutually conflicting targets. [‡]Appropriate safety refusal on unsupported pediatric profiles.

confirming real-time responsiveness for conversational deployment on WhatsApp.

Ablation Analysis. Table 8 reports an ablation study isolating the contribution of each safety and cultural component. Removing the Plan Verifier causes energy drift to jump from 2.45% to 22.40% and introduces 0.62 contraindication violations per plan. Omitting cooking oil accounting introduces a systematic 16.80% caloric underestimation. Crucially, removing the Food Compatibility Matrix preserves mathematical caloric balance (drift: 2.45%) but causes culinary realism to collapse from 9.27/10 to 5.15/10, demonstrating that clinical safety without cultural compatibility produces mathematically sound but practically unusable dietary plans.

6.3. Detailed Edge-Case Case Studies

The necessity of deterministic verification is highlighted by two critical edge cases in the benchmark manifest:

1. **SCN_007 (CKD Stage 3 with Hyper-Protein Bodybuilding):** The user requested a 2,800 kcal bodybuilding diet with 150 g/day of protein while presenting with diagnosed Stage 3 CKD (maximum safe limit: 45–65 g/day). Generative LLMs attempted to satisfy both requirements by generating an unconstrained high-protein meat diet, which in a real patient could precipitate uremic crisis and accelerate glomerular sclerosis. In contrast, DESIDIET’s portion solver detected that the constraint set $\mathcal{P}_{\text{nut}} = \{x \mid \sum x_i \text{Prot}_i \geq 150 \wedge \sum x_i \text{Prot}_i \leq 65\}$ was empty, gracefully abstaining and outputting a structured medical explanation.

Table 6: Cross-Lingual Linguistic Robustness Evaluation across Input Modalities ($N = 13$).

Linguistic Modality / Input Format	Entity Prec. (%)	Entity Rec. (%)	F1-Score	Mean Drift (%)	Pass Rate (%)	Mean Latency (s)
English (Standard Clinical Prompting)	96.4	94.8	0.956	2.31 ± 0.45	100.0	28.42 ± 3.12
Native Bengali (Unicode Script)	94.8	93.2	0.940	2.24 ± 0.52	100.0	29.10 ± 3.85
Code-Mixed <i>Banglish</i> (Latin Phonetics)	92.1	89.5	0.908	2.68 ± 0.61	90.0	29.85 ± 4.20
Macro Average / Robustness	94.4	92.5	0.935	2.41 ± 0.53	96.7	29.12 ± 3.72

Table 7: Nutritional & Clinical Guideline Conformance Matrix Across Major Pathologies in DESIDIET.

Clinical Cohort	Tier	Carb (%)	Prot (g)	Fat (%)	Fiber (g)	Na (mg)	K (mg)	Fe (mg)	Ca (mg)
Healthy Adult Male	Target DESIDIET	55–65% 58.2%	54.0 56.4	20–30% 24.1%	≥ 30.0 32.4	$\leq 2,300$ 1,940	$\geq 3,500$ 3,620	≥ 19.0 21.2	$\geq 1,000$ 1,040
Anemia Risk Female	Target DESIDIET	55–65% 57.4%	46.0 48.2	20–30% 23.5%	≥ 25.0 27.1	$\leq 2,000$ 1,780	$\geq 3,500$ 3,540	≥ 29.0 29.8	$\geq 1,000$ 1,080
Type 2 Diabetes Male	Target DESIDIET	45–55% 49.1%	60–75 68.5	20–25% 22.8%	≥ 35.0 36.8	$\leq 2,000$ 1,650	$\geq 3,500$ 3,590	≥ 19.0 20.4	$\geq 1,000$ 1,015
Hypertension Female	Target DESIDIET	50–60% 53.6%	46.0 47.8	20–25% 21.4%	≥ 25.0 28.5	$\leq 1,500$ 1,380	$\geq 3,500$ 3,680	≥ 29.0 29.4	$\geq 1,000$ 1,030
CKD Stage 3 Male	Target	55–65%	≤ 45.0	20–30%	≥ 25.0	$\leq 1,800$	$\leq 2,000$	≥ 19.0	800– 1,000
	DESIDIET	61.2%	43.2	22.1%	26.2	1,520	1,840	19.5	890
Gestational 2nd Trim.	Target	55–65%	≥ 61.0	20–30%	≥ 28.0	$\leq 2,000$	$\geq 3,500$	≥ 38.0	$\geq 1,200$
	DESIDIET	56.8%	64.5	25.2%	30.1	1,820	3,710	38.9	1,240

Table 8: Ablation Analysis of Pipeline Safety and Cultural Controls.

Ablation Variant	Violations	Drift (%)	Culinary
Full DesiDiet Pipeline	0.00	2.45	9.27 / 10
w/o Plan Verifier	0.62	22.40	9.15 / 10
w/o Portion Optimizer	1.15	14.82	6.85 / 10
w/o Compatibility Matrix	0.00	2.45	5.15 / 10
w/o Safety Guard	0.23	2.45	9.25 / 10
w/o Cooking Oil Model	0.00	16.80	7.20 / 10

2. **SCN_008 (Pediatric Infant Feeding):** A profile specifying an infant aged 1.0 year requesting adult complementary feeding was routed to the pre-routing safety classifier. The classifier immediately rejected the query with a pediatric referral, preventing the application of adult dietary algorithms to early childhood.

7. Real-World Deployment & Usability Artifacts

7.1. Multi-Channel Accessibility

User attrition in digital health applications frequently exceeds 75% within 30 days due to manual data logging friction [21]. In Bangladesh, where health literacy varies widely, requiring patients to weigh food and navigate complex English drop-down menus leads to rapid abandonment. DESIDIET addresses this by deploying across three synchronized modalities:

1. **Web Clinical Dashboard:** Provides structured weekly meal schedules, macronutrient progress indica-

tors, and micronutrient tracking against reference intake envelopes.

2. **Native Bengali WhatsApp Interface:** Enables patients to query dietary recommendations and log meals via natural conversational Bengali without requiring the installation of standalone applications.
3. **Deterministic Portion Verification Engine:** Displays transparent nutrient recalculations alongside validated portion allocations and contraindication-safe alternatives in real time.

7.2. Bilingual Conversational Interaction

Below is an authentic trace of a patient consultation over the WhatsApp interface:

Patient: “Amar diabetes ache ar pressure o high. Dupure ki khete pari?” [I have diabetes and high blood pressure. What can I eat for lunch?]

DesiDiet (Pusti AI): “Apanar diabetes o uccho roktocaper jonno dupurer ekti sushomo o nirapod khabarel talika niche dewa holo (lobon o chini chara prostutkrito):

1. Lal chaler bhat: 1 bati (150 g) — jotil shorkora o fiber shomriddho
2. Rui macher patla jhol: 1 tukra (100 g), 10 g shorishar tele ranna
3. Palang shak bhaji: 1 bati (100 g) — potassium o vitamin shomriddho
4. Masur dal: adha bati (50 g)

Pushtiman: Shakti 580kcal, Carbohydrate 68 g, Protein 29 g, Sodium 380 mg. Otirikto lobon porihar korben.”

[Translation: For your diabetes and hypertension, here is a balanced and safe lunch plan (prepared without extra salt or sugar): 1. Brown rice: 1 bowl (150 g); 2. Rohu fish curry: 1 piece (100 g) cooked in 10 g mustard oil; 3. Spinach bhaji: 1 bowl (100 g); 4. Lentil dal: 0.5 bowl (50 g). Nutritional summary: Energy 580 kcal, Carbs 68 g, Protein 29 g, Sodium 380 mg. Avoid raw added salt.]

8. Discussion & Research Governance

8.1. Verified Software Evidence vs. Clinical Trial Claims

In compliance with scientific integrity standards, we clearly demarcate the boundaries of our findings:

1. **Unit and Integration Verification:** All 101 tests across the backend test suite (`backend/tests/`) pass with 100% success, verifying the correctness of data normalization, portion mathematics, disease matching, and plan verification.
2. **Decision-Support Boundary:** DESIDIET is engineered as an automated clinical decision-support prototype for informational assistance. It is not an autonomous medical prescribing device.

3. **Absence of Clinical Trials:** We explicitly disclaim claims of clinical trial efficacy. Demonstrating improvements in patient glycemic control (HbA1c) or renal biomarkers requires prospective, longitudinal clinical trials overseen by medical ethical committees.

8.2. Mitigating Cognitive Friction in South Asian Users

User attrition in digital health apps frequently exceeds 75% within 30 days due to manual data logging friction [21]. In Bangladesh, where health literacy varies widely, requiring patients to weigh food and navigate complex English drop-down menus leads to rapid attrition. DESIDIET addresses this by mapping gram portions to intuitive Bengali household units (“1 bati” [1 bowl], “1 tukra” [1 piece], “1 chamoch” [1 tablespoon]) and offering native Bengali consultation over WhatsApp, lowering barriers for low-literacy populations.

9. Limitations & Future Work

Key limitations include:

1. **Dataset Currency:** The Bangladesh FCT dates from 2014; recent soil mineral depletion and commercial fortification are not fully captured.
2. **Cooking Retention Tables:** Raw-to-cooked nutrient retention currently relies on standard USDA retention tables; dedicated Bangladeshi cooking retention coefficients are needed.
3. **Prospective Medical Trials:** Longitudinal patient studies are required to evaluate adherence and health outcomes.

10. Conclusion

This paper has presented an artifact-grounded systems evaluation of DESIDIET, a culturally tailored knowledge graph and deterministic portion-planning framework for South Asian nutrition. By replacing scale-invariant cosine heuristics with an exact linear portion optimization solver, incorporating a curated 575-item Food Compatibility Matrix for culturally coherent meal structures, and introducing an independent post-generation Plan Verifier, DESIDIET resolves the critical vulnerabilities of portion hallucination, caloric drift, contraindication failure, and bizarre food pairings in generative nutrition AI. Grounded in 582 canonical Bangladeshi foods, national dietary guidelines, and authentic culinary roles, the system constrains caloric drift to 2.45%, eliminates clinical contraindications, and elevates culinary realism to 9.27/10, establishing a reproducible foundation for safe, scalable, and meaningful dietary decision support.

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Ethics and Data Availability

DESIDIET is a decision-support prototype. No human patient data was utilized during benchmark testing; all scenarios were synthetically generated. Source code and benchmarks are available at: <https://github.com/LegendaryBeast/desi-diet-AI>.

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